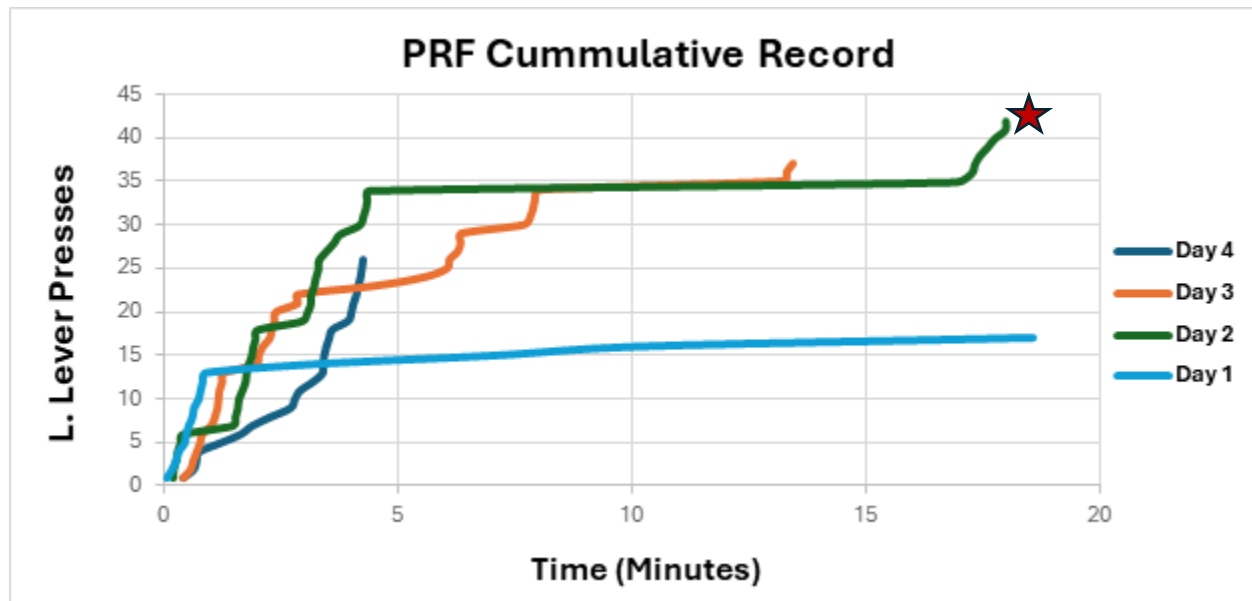
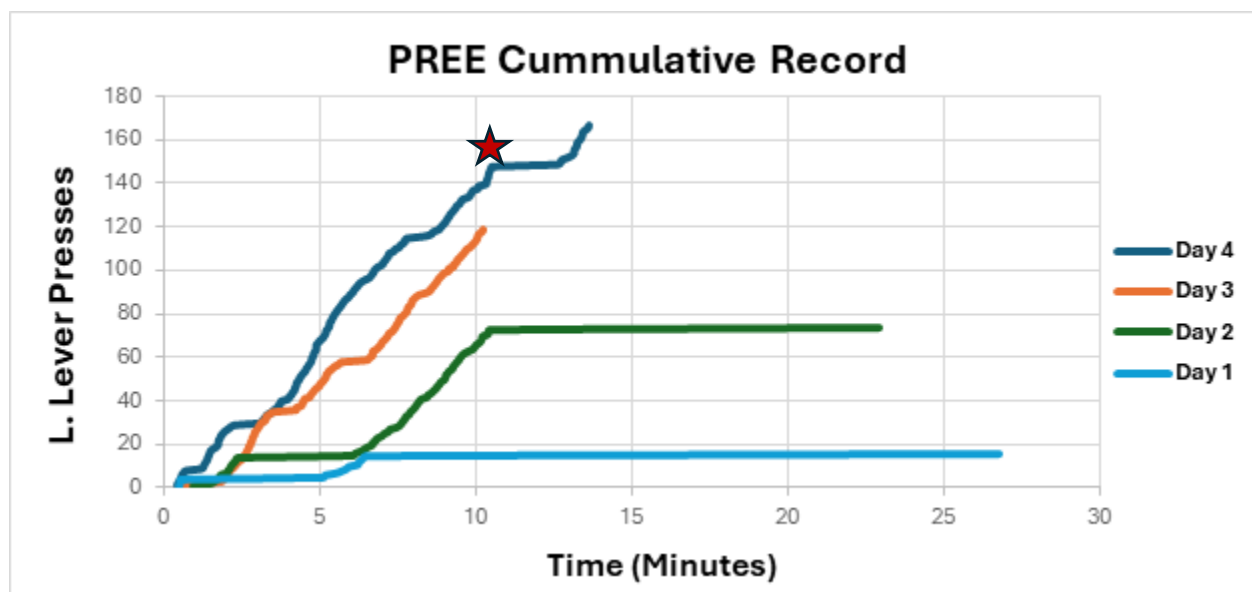


Assignment #3: Partial Reinforcement and Partial Reinforcement Extinction Effect

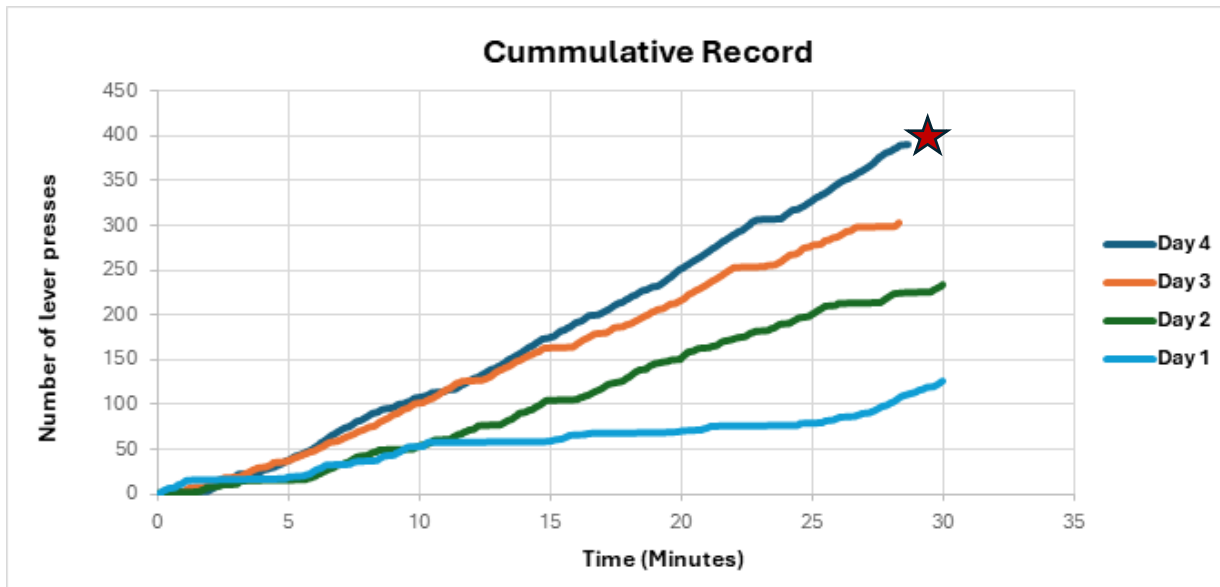
Experiment #6 graph:



Experiment #7 graph:

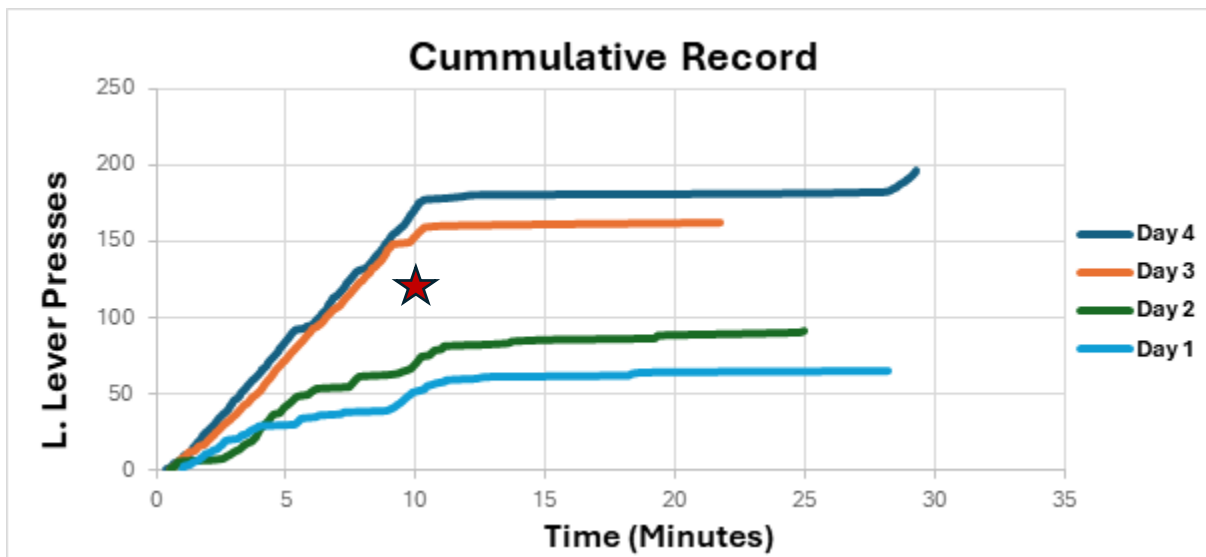


Short-answer questions:



- Compare and contrast the results you saw between Experiment 4 and Experiment 6. How did your animal respond differently to a partial reinforcement schedule than a continuously reinforced schedule? Explain.

Comparing the results from experiment #4 (seen above) and experiment #6 (on page 1), there are two major discussion points. The first is a similarity in the results of both experiments, which is that the expected “pattern of response” was successfully and appropriately expressed (smooth/straight line for CRF and staggered/varying line for PRF). The other major comparison is a difference in the data sets, the difference being a significant decline in the “rate of response” or “slope” from experiment #4 to experiment #6, with the maximum for the slope (being number of lever presses) decreasing significantly (from ~390 to ~42 presses). In conclusion, being trained on a partial reinforcement schedule, my rat’s response pattern was different from that of a continuously reinforced schedule, but ALSO the responses differed from what was expected (of using a PRF schedule) since my rat quit responding prematurely (causing the plateau’s seen in the data and the abrupt end in slope for day 4 of the experiment #6 graph). This indicates “ratio strain,” probably due to “leaning-out” the reinforcement schedule too quickly, which meant that likely very little or zero “partial reinforcement extinction effect” was to be expected of my mouse in experiment #7.



- Compare and contrast the results from Experiments 5 and Experiment 7. Describe the differences and similarities that are visible on your graphs and give explanations for those differences. Explain.

Again, we see another successful training with experiment #5 as all the lines in this graph (seen just above) are relatively straight until all four lines flatten out at the 10-minute mark when the extinction period began. The same statement is true for days 1-3 of the experiment #7 data indicating that no “resistance to extinction” had formed. However, on day 4 of experiment #7, thankfully we can see a difference in my rat’s response when, after 10-minute mark, my rat continued to press the lever ~3 minutes into the extinction period (past the point where the other days had plateaued in their slope). This indicates that “partial reinforcement extinction effect” WAS present for day 4 of the collected data, which in turn indicates some level of success in my partial reinforcement conditioning process. In conclusion, the results from experiment #5 and #7 are very similar, although the single key difference (represented as the star on #7’s graph) is essential in explaining how partial reinforcement schedules are different from continuous reinforcement schedules, their effects an organism, AND what “partial reinforcement extinction effects” even are in the first place. The reason we see this difference in the data ONLY for “day 4” of experiment #7 and not the other days is very likely due to the ratio-strain that I discussed for days 1-3 of experiment #6 (in my previous response).